Using LLM for Improving Mapping between Enterprise Architectural Views

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***Abstract*—This paper presents an AI-assisted semantic traceability workflow for recommending candidate mappings between DoDAF architectural viewpoints. The workflow uses dense semantic embeddings for initial candidate retrieval through cosine similarity, followed by cross-encoder re-ranking to refine candidate ordering. Candidate mappings are validated by a subject matter expert (SME) and evaluated based on candidate coverage, recommendation ranking, and relevance within a top-3 review budget. The approach was evaluated on a real-world Positioning, Navigation, and Timing (PNT) Enterprise Architecture developed for a U.S. federal government department. Three mapping tasks were examined: Capability-to-Operational Activity, Operational Activity-to-Service, and Capability-to-Service. At least one SME-validated relationship appeared within the first three recommendations for 100%, 95%, and 100% of the evaluated cases, respectively. Cross-encoder re-ranking improved the placement of valid relationships for Operational Activity-to-Service mapping, but the observed differences from cosine retrieval were not statistically significant. A stricter relevance criterion reduced the proportion of fully relevant recommendations, highlighting the need for SME validation to distinguish complete from partial traceability relationships. The findings support semantic retrieval as a decision-support mechanism for cross-viewpoint traceability while motivating further evaluation across additional architectures, viewpoints, and domains.**

***Index Terms*—component, formatting, style, styling, insert**

***Index Terms*—LLM, LLM-as-a-judge, Requirement Engineering, CONOPS**

I. Introduction

Modern enterprise systems engineering relies heavily on standardized architecture frameworks, such as the Department of Defense Architecture Framework (DoDAF) and the Unified Architecture Framework (UAF), to model complex, multi-domain systems. A core tenet of these frameworks is the establishment of rigorous, cross-layer traceability between architectural viewpoints. For instance, DoDAF advocates mapping Strategic Capabilities to Operational Activities to ensure enterprise alignment, as well as mapping Operational Activities to services, systems, software, and components to verify that physical and logical solutions fulfill required operational functions. Establishing this inter-viewpoint alignment guarantees that high-level strategic goals directly inform operational behaviors and downstream resource implementations.

Despite its critical importance, manually establishing and maintaining inter-viewpoint traceability matrices across large-scale enterprise architectures is extraordinarily labor-intensive and costly. Systems engineers frequently spend hundreds of hours facilitating intensive review workshops with Subject Matter Experts (SMEs) to manually validate these mappings. As architectures expand or undergo iterative modifications, manual tracing becomes increasingly susceptible to human error, resulting in missing traceability links, unfulfilled operational activities, or orphaned system resources.

To mitigate this bottleneck, recent developments in Large Language Models (LLMs) and Information Retrieval (IR) offer a promising avenue for improving mapping between architectural views. This paper presents an AI-assisted workflow designed to automatically discover, rank, and recommend candidate traceability links between DoDAF architectural viewpoints, serving as an automated decision-support tool for systems engineers. The workflow leverages dense semantic embeddings generated from model element textual descriptions to conduct initial candidate retrieval (*rank-by-cosine*), followed by a cross-encoder re-ranking stage (*rank-by-rerank*) to capture nuanced semantic relationships between source and target model elements.

To evaluate the effectiveness and reliability of the generated mappings, the proposed workflow was applied to an Position, Navigation, and Time (PNT) enterprise architecture , modeled in DoDAF using Cameo Systems Modeler. The automated recommendations were validated by a SME. We demonstrated that the workflow recovers complex architectural relationships, potentially reducing engineer inspection effort. While demonstrated on DoDAF models exported from Cameo, the underlying workflow is tool- and framework-agnostic, making it broadly applicable across Enterprise Architecture (EA) and Model-Based Systems Engineering (MBSE) domains.

The primary contributions of this paper are as follows:

- **Automated Inter-Viewpoint Mapping Workflow:** We

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introduce a two-stage retrieval-then-reranking semantic workflow tailored specifically for establishing traceability across DoDAF Capability, Operational, and Service viewpoints.
- **Industrial Case Study Validation:** We demonstrate the practical utility of the workflow on a real-world, industrial-scale DoDAF enterprise architecture developed for a U.S. federal government department in Cameo Systems Modeler.
- **Rigorous SME Ground Truth Evaluation:** A SME validated the mappings using rank-aware Information Retrieval metrics to measure retrieval yield and engineering review efficiency.

The remainder of this paper is organized as follows: Section II Related Work reviews related literature in AI-driven traceability link recovery and synthesizes current research gaps. Section III Technical Approach details the case study, a six-step semantic traceability workflow used to automatically provide link candidates between DoDAF viewpoints, and the artificial intelligence platform (AIP) used to support this workflow. Section IV Results presents the empirical evaluation results, followed by conclusions and future work.

II. Related Work

A. Evaluation Approach

To systematically assess AI-driven approaches for establishing links between DoDAF architectural viewpoints, we synthesize an evaluation framework anchored in Information Retrieval for Traceability Link Recovery (IR-TLR) protocols [1], [2] and re-ranking paradigms.

Our comparative framework evaluates existing literature across four core dimensions:

1) **AI & Retrieval Architecture:** Categorizes the underlying retrieval and ranking mechanisms—ranging from classic Vector Space Models (VSM/LSI) and bi-encoder Transformer embeddings (e.g., BERT, SimCSE) to two-stage retrieval-then-reranking pipelines (cross-encoders and LLM re-rankers).
2) **Link Quality & Top-k IR Rigor:** Evaluates candidate recommendation quality as a top-k decision-support task for systems engineers using rank-aware Information Retrieval metrics (evaluated at a fixed review budget, e.g., $k = 3$).
3) **Architectural Interpretability & Governance:** Examines the transparency of generated candidate links, evaluating whether the approach provides scalar similarity scores or actionable semantic justifications suitable for engineering review and model audits.
4) **Empirical Validation & MBSE Toolchain Integration:** Evaluates benchmark realism, contrasting studies validated on synthetic or static textual documents against those using formal MBSE repository models (e.g., Cameo Systems Modeler exports) across interconnected DoDAF views.

B. Background

The establishment and maintenance of traceability links across architectural viewpoints are essential for ensuring operational coverage, resource justification, and capability fulfillment. Traditionally, maintaining consistency across DoDAF or UAF viewpoints relied on manual inspection by SMEs. Recent advances in Natural Language Processing (NLP), dense semantic embeddings, and Large Language Models (LLMs) offer automated alternatives for model-to-model alignment.

1) Transition from Keyword IR to Dense Semantic Embeddings: Early automated traceability approaches relied heavily on statistical IR methods such as Term Frequency-Inverse Document Frequency (TF-IDF) and Latent Semantic Indexing (LSI) [3]. While efficient, these methods struggle with vocabulary mismatch across DoDAF viewpoints: Strategic capabilities are often expressed in high-level business terminology, whereas Operational Activities and Services components employ technical or domain-specific language.

To bridge this semantic divide, domain-adapted Transformer architectures were introduced. Lin et al. [4] demonstrated that fine-tuning pre-trained language models via contrastive learning captures contextual semantics far superior to keyword-matching baselines. Bonner et al. [5] extended transformer embeddings to MBSE environments, establishing semantic links between high-level automotive functional descriptions and AUTOSAR/SysML design elements. Their work demonstrated that embedding model metadata alongside textual descriptions significantly reduces false-positive link suggestions in structured system models.

2) Two-Stage Retrieval and Cross-Encoder Re-ranking: While bi-encoder embedding models (such as BERT or Sentence-Transformers) enable rapid candidate retrieval via vector space similarity, they process source and target artifacts independently. This independent encoding can fail to capture fine-grained semantic dependencies between distinct DoDAF entity types, such as an Operational Activity and a supporting System Resource.

Information Retrieval research has increasingly adopted two-stage retrieval-then-reranking architectures to resolve this limitation [6], [7]. In the first stage, bi-encoder embeddings retrieve a broad candidate list based on cosine similarity (*rank-by-cosine*). In the second stage, a cross-encoder jointly processes candidate pairs (e.g., Capability vs. Operational Activity), performing deep cross-attention across both textual descriptions to refine candidate ordering (*rank-by-rerank*). Applying two-stage re-ranking to MBSE model element pairs provides an efficient mechanism for prioritizing candidate traceability links prior to human engineering review.

3) LLM-Assisted Model Alignment and Semantic Recovery: With the advent of generative LLMs, researchers have explored LLMs for automated architectural analysis and entity relationship extraction. Fuchß et al. [8] introduced *ArTEMiS*, demonstrating that LLMs can extract architectural entities and candidate relations from unstructured technical documentation. Similarly, Niu et al. [9] applied Retrieval-Augmented Gener-

ation (RAG) to inject architectural context into LLM prompts to assist in link identification.

However, existing generative LLM approaches often treat traceability recovery as a binary classification task or free-form text generation. In formal MBSE frameworks, systems engineers require a prioritized, rank-ordered candidate list constrained by standard IR metrics rather than unconstrained text outputs.

C. Comparative Evaluation of Existing Approaches

Evaluating the current literature against our MBSE assessment framework reveals key strengths and ongoing limitations in automated inter-viewpoint mapping:

1) Criterion 1: AI & Retrieval Architecture: Dense vector retrieval systems [4], [5] provide high scalability for candidate generation but rely solely on bi-encoder cosine similarity. These single-stage models struggle to differentiate complex typed relationships inherent in DoDAF/UAF viewpoints (e.g., distinguishing whether a System Resource *performs*, *hosts*, or merely *supports* an Operational Activity). While RAG and LLM-assisted pipelines [8], [9] capture richer contextual nuances, they rarely incorporate cross-encoder re-ranking specifically tuned to compare structural model element descriptions exported from MBSE repositories.

2) Criterion 2: Link Quality & Top-k IR Rigor: Evaluating link recovery as a top-k recommendation task is critical for practical systems engineering, as domain experts require prioritized candidate lists to minimize manual validation overhead. Fine-tuned embedding studies (Lin et al. [4], Bonner et al. [5]) demonstrate strong IR evaluation rigor, consistently reporting rank-aware metrics such as Precision@k, Recall@k, and MAP. In contrast, generative LLM studies (e.g., Eisenreich et al. [10], Fuchß et al. [8]) frequently report aggregate binary classification metrics (F_1-score, unranked Precision/Recall). By treating link identification as a binary decision rather than a ranked recommendation stack, these approaches fail to quantify human inspection effort across realistic review budgets (e.g., $k = 3$).

3) Criterion 3: Architectural Interpretability & Governance: Classical embedding models operate as numerical black boxes, outputting similarity scores without semantic explanation. Generative LLM architectures improve auditability by generating step-by-step reasoning or justifications for suggested links. However, for formal UAF/DoDAF viewpoint mapping, deterministic top-k candidate ordering verified against SME ground truth remains the primary requirement for airworthiness and enterprise certification audits.

4) Criterion 4: Empirical Validation & MBSE Toolchain Integration: A major limitation of existing TLR benchmarks is their heavy reliance on open-source software repositories (e.g., Java codebases) or unstructured textual documents. These benchmarks do not reflect the formal metamodel structures, concise model element descriptions, or inter-viewpoint dependency networks found in industrial MBSE environments. Few studies evaluate traceability directly on model elements extracted from industrial-based repositories containing actual UAF/DoDAF viewpoints.

D. Summary of Literature Gaps

Our review identifies three primary gaps in the current literature regarding automated UAF/DoDAF viewpoint mapping:

1) **Focus Disparity (Requirements vs. Inter-Viewpoint Mapping):** Prior TLR research focuses predominantly on requirements-to-code or requirements-to-model recovery. Automated semantic mapping *between UAF/DoDAF viewpoints* within MBSE repositories remains underexplored.
2) **Lack of Two-Stage Re-ranking in MBSE Traceability:** Most MBSE link retrieval tools rely exclusively on single-stage vector embeddings. The combination of dense embedding retrieval followed by cross-encoder semantic re-ranking has not been systematically evaluated on model elements defined in different viewpoints.
3) **Absence of Rank-Aware Metric Evaluation for Alignment:** Existing LLM-based architectural mapping studies often lack rank-aware IR evaluations. There is a need for empirical benchmarks measuring Precision@k, Recall@k, MAP, and F_1@k at realistic engineering review cutoffs ($k = 3$) against SME-established ground truth.

III. Technical Approach

A. The Positioning, Navigation, and Timing (PNT) Enterprise Architecture

Positioning, Navigation, and Timing (PNT) data constitute a foundational utility for modern critical infrastructure, enabling essential operations across transportation, telecommunications, energy distribution, finance, and defense. Over the past two decades, numerous public- and private-sector organizations have developed PNT architectures, strategic blueprints, and System-of-Systems (SoS) methodologies to address PNT dependencies, mitigate single points of failure—including unmitigated reliance on space-based signals—and coordinate technology roadmaps across multiple domains [11]–[17].

To evaluate the proposed LLM-driven mapping approach, this study applies the workflow to a PNT Enterprise Architecture (EA) developed for a U.S. federal government department[1]. The PNT EA was defined using DoDAF architectural viewpoints, including the Capability, Operational, and Service viewpoints. These three viewpoints provide the structured basis for assessing whether the LLM-assisted workflow can help systems engineers establish cross-viewpoint relationships with sufficient semantic fidelity to support enterprise-scale traceability analysis. The DoDAF-based PNT EA comprises 39 Strategic Capabilities, 80 Operational Activities, and 36 Services, represented within the respective Capability, Operational, and Service viewpoints.

[1]Due to confidentiality constraints, the department name and the complete EA models cannot be disclosed.

By establishing semantic mappings across these three DoDAF viewpoints, this study provides an end-to-end evaluation framework for assessing the LLM's ability to connect high-level strategic capabilities with operational activities and service-level representations.

B. The Artificial Intelligence Platform

The experiments reported in this study were conducted on a secure AIP (Artificial Intelligence Platform). The platform is designed to operate within a protected environment with no internet connectivity, preventing data from being shared with external networks and safeguarding sensitive information from unauthorized exposure. The AIP provides the following models for semantic retrieval and ranking that were used in this study: **llama-embed-nemotron** [2]; **llama-nemotron-rerank** [3]. A comparative evaluation of embedding and re-ranking models is beyond the scope of this paper. Readers are referred to [18], [19] for comparisons of embedding models and to [20] for comparisons of re-ranking models.

C. AI-Assisted Semantic Traceability Workflow

As illustrated in Figure 1, the proposed workflow comprises six stages that support the establishment of candidate traceability links between UAF/DoDAF architectural viewpoints. The workflow transforms these viewpoints into dense semantic embeddings. Embedding-based retrieval and cross-encoder reranking are then used to identify and prioritize candidate traceability links. Validation activities are incorporated throughout the workflow, enabling a systematic assessment of both computational behavior and traceability effectiveness within a unified methodological framework.

a) Step 1 – Extraction of DoDAF Architectural viewpoints: The workflow begins by exporting architectural viewpoints from the Cameo Systems Modeler repository. In this case study, the following DoDAF viewpoints were addressed:

- Capability viewpoint containing enterprise capabilities, their interdependencies, and business drivers.
- Operational viewpoint containing the enterprise operational activities and performers.
- Service viewpoint containing the services and their interconnections supporting or providing the PNT functions.

DoDAF obviously contains many other viewpoints. However, the case study analyzed in this report at the time this study was conducted had a limited set of DoDAF viewpoints defined.

b) Step 2 – Semantic Embedding Generation: Model elements along their documentation exported from Cameo are converted into dense semantic vector representations using the AIP's embedding model. Each model element, such as a capability, is represented by a single embedding vector.

c) Step 3 – Candidate Trace Retrieval: For each source model element, e.g., capability, cosine similarity is computed between its embedding vector and the embeddings of all target model elements, e.g., operational activities. This produces an initial ranking of traceability candidates, referred to as *rank-by-cosine*.

d) Step 4 – Cross-Encoder Re-ranking: To refine candidate prioritization, the top-3 DoDAF architectural viewpoints retrieved by the embedding-based retrieval stage are re-evaluated using a cross-encoder. Whereas embedding models represent artifacts independently, the cross-encoder jointly analyzes pairs of potential mapping, such as capability vs operational activity, and so on, enabling a more comprehensive assessment of semantic relevance. The resulting ranking, denoted *rank-by-rerank*, improves the ordering of semantically similar candidates while retaining the efficiency of embedding-based retrieval. As reported by [6], [7], retrieval-then-reranking architectures have become increasingly common in information retrieval systems because they enable efficient candidate retrieval followed by a more detailed relevance assessment.

e) Step 5 – Subject Matter Expert Validation: The ranked candidate traceability links are reviewed by a SME. For each proposed link, the SME examines the source and target architectural elements, along with any available supporting documentation, to assess the validity of the cross-viewpoint relationship. For example, when mapping capabilities to operational activities, the SME evaluates whether each candidate activity supports the source capability. This validation approach is applied consistently across all mapping types evaluated in the study. Each candidate link is classified as:

- **Accepted**: the target element adequately supports or realizes the intent of the source element.
- **Partially Accepted**: the target element supports or realizes only part of the intent of the source element.
- **Rejected**: the target element does not provide a meaningful traceability relationship to the source element.

The resulting SME judgments constitute the reference traceability baseline used to evaluate the candidate rankings and the effectiveness of the proposed workflow.

f) Step 6 – Retrieval Performance Evaluation: The effectiveness of the workflow is assessed by comparing the automatically generated rankings against the SME baseline using standard Information Retrieval (IR) metrics as defined in Table I.

IV. Results

This study empirically evaluates the AI-assisted semantic workflow for mapping relationships across DoDAF viewpoints: Capability-to-Operational Activity, Operational Activity-to-Service, and Capability-to-Service. To do so, performance is evaluated across two distinct relevance scoring criteria:

- **Rule 1 (1-1-0):** An inclusive binary relevance rule where both *Accepted* and *Partially Accepted* links are treated as relevant (1), and *Rejected* links are treated as non-relevant (0).
- **Rule 2 (2-1-0):** A strict/weighted relevance rule where *Accepted* links receive a weight of 2, *Partially Accepted* links receive 1, and *Rejected* links receive 0, evaluating

[2]https://huggingface.co/nvidia/llama-embed-nemotron-8b

[3]https://huggingface.co/nvidia/llama-nemotron-rerank-1b-v2

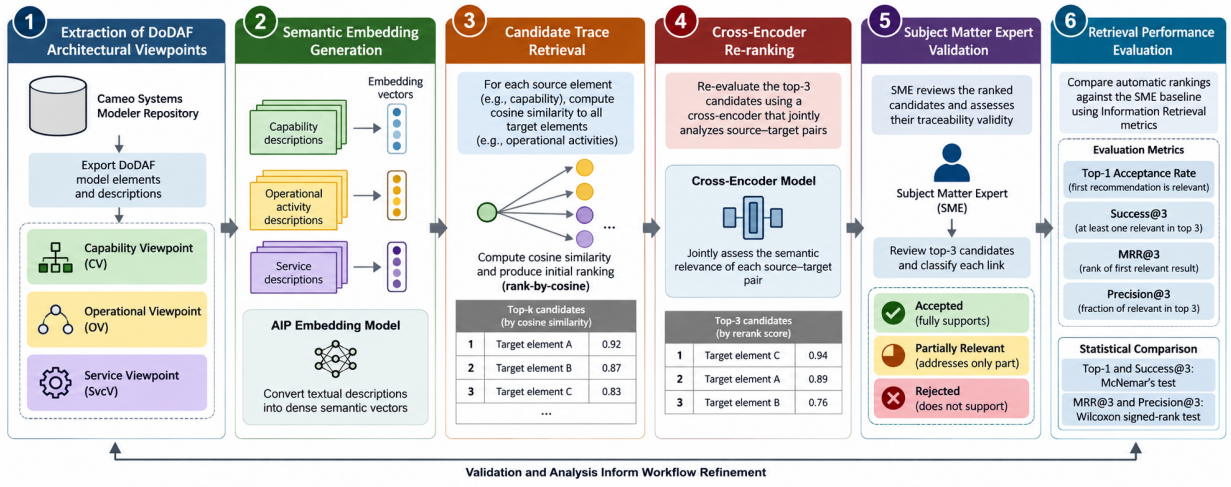


Fig. 1. AI-Assisted Semantic Traceability Workflow

TABLE I
INFORMATION RETRIEVAL EVALUATION METRICS.

Metric	Definition
Top-1 Acceptance Rate	How often the workflow's first (highest-ranked) recommendation is accepted by the SME as a valid traceability link.
Success@3	How often at least one valid traceability link appears somewhere among the workflow's top three recommendations.
MRR@3	Measures how close to the top of the first three recommendations the first valid traceability link appears. A valid result ranked first receives more credit than one ranked second or third.
Precision@3	Measures how many of the workflow's top three recommendations are judged relevant by the SME.

the concentration of fully realized traceability relationships.

Table II summarizes the retrieval metrics across all three architectural mapping tasks, candidate retrieval strategies, and relevance scoring rules.

A. Retrieval Performance Across Architectural Mappings

Across all evaluated viewpoints, the semantic retrieval workflow consistently surfaces relevant target model elements within a compact review budget ($k = 3$):

- **Capability-to-Operational Activity (CV-6):** Achieving complete candidate coverage (Success@3 = 1.000) guarantees a zero-miss baseline within a 3-item review budget, eliminating orphaned capabilities prior to manual inspection. Although a high inclusive Precision@3 (0.897) confirms that the model accurately locates the correct operational neighborhood, the drop under weighted evaluation (0.731) highlights the gap between semantic affinity and strict functional allocation. While AI retrieval successfully aggregates the relevant functional cluster, SME validation remains necessary to distinguish primary realization from supporting or partial coverage.
- **Operational Activity-to-Service (SvcV-5):** Tracing abstract operational workflows (OV-5a) to technical service implementations (SvcV-1) yields high candidate recovery (Success@3 = 0.950)[cite: 16]. Unlike capability-level mappings, Cross-Encoder Reranking provides a distinct advantage here, increasing the Top-1 Acceptance Rate from 0.863 to 0.900 and MRR@3 from 0.902 to 0.925. Because services rely on specialized interface terminology, the cross-encoder's deep attention mechanism effectively resolves cross-viewpoint vocabulary mismatches to place the primary service first 90% of the time. Furthermore, the Precision@3 drop from Rule 1 (0.725) to Rule 2 (0.477) reflects service-oriented modularity: individual services rarely fulfill an operational activity in isolation, requiring SME verification of service composability and orchestration.
- **Capability-to-Service (CV-7):** Directly mapping strategic objectives to technical implementations also achieves perfect candidate recovery (Success@3 = 1.000). Tracing across such wide abstraction gaps favors single-stage Cosine retrieval, which outperforms Cross-Encoder Reranking at Top-1 (0.872 vs. 0.846) and MRR@3 (0.932 vs. 0.910) by capturing global semantic intent without over-indexing on technical wording discrepancies. The margin between inclusive Precision@3 (0.803) and weighted Precision@3 (0.504) reflects the macro-level nature of capabilities, which demand an ecosystem of complementary services rather than a single monolithic

TABLE II
RETRIEVAL METRICS ACROSS INPUT TYPES AND RELEVANCE SCORING RULES.

Input	Relevance Rule	Method	Top-1 Acceptance Rate	Success@3	MRR@3	Precision@3
CV-6	Rule 1 (1-1-0)	Cosine	0.846	1.000	0.919	0.897
		Rerank	0.821	1.000	0.910	0.897
	Rule 2 (2-1-0)	Cosine	0.846	1.000	0.919	0.731
		Rerank	0.821	1.000	0.910	0.731
SvcV-5	Rule 1 (1-1-0)	Cosine	0.863	0.950	0.902	0.725
		Rerank	0.900	0.950	0.925	0.725
	Rule 2 (2-1-0)	Cosine	0.863	0.950	0.902	0.477
		Rerank	0.900	0.950	0.925	0.477
CV-7	Rule 1 (1-1-0)	Cosine	0.872	1.000	0.932	0.803
		Rerank	0.846	1.000	0.910	0.803
	Rule 2 (2-1-0)	Cosine	0.872	1.000	0.932	0.504
		Rerank	0.846	1.000	0.910	0.504

asset. Here, SME oversight ensures that the retrieved service portfolio provides collective realization.

B. Strategy Comparison: Cosine vs. Cross-Encoder Reranking

To determine whether the reranking provides a statistically meaningful improvement over single-stage bi-encoder retrieval, paired hypothesis tests were performed for each metric across input types and relevance rules. Top-1 Acceptance Rate and Success@3 were evaluated using McNemar's test for paired nominal data, while MRR@3 and Precision@3 were evaluated using the Wilcoxon signed-rank test for paired continuous/ordinal distributions.

The paired comparisons confirm that Cosine similarity and Cross-Encoder Reranking exhibit statistically indistinguishable performance across all tasks and metrics ($p > 0.05$):

- For operational-to-service mapping, reranking increases Top-1 Acceptance Rate by 3.7% (0.863 to 0.900) and MRR@3 by 0.023 (0.902 to 0.925), though these improvements do not reach statistical significance ($p = 0.4531$ and $p = 0.2500$, respectively).
- For capability-based mappings, Cosine similarity slightly outperforms Reranking at Top-1 (0.846 vs. 0.821 for `capability-to-operational activity`; 0.872 vs. 0.846 for `capability-to-service`), but these differences are statistically non-significant ($p = 1.0000$).
- Precision@3 and Success@3 values are identical between Cosine and Rerank across all inputs and rules ($p = 1.0000$). Because the cross-encoder only reorders the top three candidate items retrieved during the initial vector search, candidate set membership remains unchanged, resulting in identical set-based precision and recall boundaries.

C. Impact of Relevance Scoring Rules

To evaluate how strict acceptance criteria affect engineering review quality, Precision@3 was compared between Rule 1 (inclusive relevance) and Rule 2 (strict/weighted relevance) using the Wilcoxon signed-rank test.

The transition from Rule 1 to Rule 2 results in a statistically significant reduction in Precision@3 across all architectural mapping types ($p = 0.0002$):

- For Capability-to-Operational Activity, mean Precision@3 decreases from 0.897 to 0.731 ($p = 0.0002$).
- For Operational Activity-to-Service, mean Precision@3 decreases from 0.725 to 0.477 ($p = 0.0002$).
- For Capability-to-Service, mean Precision@3 decreases from 0.803 to 0.504 ($p = 0.0002$).

This significant drop demonstrates that while the workflow consistently presents candidates that are at least partially relevant (Rule 1), a substantial fraction of top-3 recommendations correspond to partial realizations or overlapping functional coverage rather than complete realization. These findings highlight the essential role of SME oversight in validating enterprise architecture traceability: while AI-driven semantic retrieval effectively identifies relevant candidate links, human inspection remains necessary to distinguish full realization from partial functional alignment.

D. Key Takeaways for Systems Engineering Practice

1) **Semantic Retrieval Can Support Initial Traceability Candidate Discovery:** Manual establishment and review of cross-viewpoint traceability can require substantial systems engineering effort, particularly as enterprise architectures become larger and more interconnected. In the evaluated PNT case study, the workflow achieved Success@3 ≥ 0.950 across all three mapping tasks, indicating that relevant candidate relationships were consistently present within the top-3 recommendations. These results suggest that semantic retrieval can support the initial discovery of traceability candidates and provide a bounded set of relationships for subsequent SME review. The study did not directly measure reductions in review time or the elimination of orphaned architectural elements; such effects therefore require further evaluation.
2) **Cross-Encoder Re-ranking Did Not Produce a Statistically Significant Overall Improvement:** In the

evaluated case study, paired statistical tests found no statistically significant differences between cosine-based retrieval and cross-encoder re-ranking across the reported metrics ($p > 0.05$). Re-ranking improved Top-1 Acceptance Rate from 0.863 to 0.900 and MRR@3 from 0.902 to 0.925 for Operational Activity-to-Service mapping, but these differences were not statistically significant. For the two capability-based mappings, cosine retrieval produced slightly higher Top-1 and MRR@3 values, also without statistically significant differences. These findings suggest that the additional re-ranking stage did not provide a statistically demonstrated advantage within the scope of this evaluation. Broader conclusions regarding computational cost or model complexity require additional measurements and experiments.

3) **SME Validation Remains Important for Distinguishing Full and Partial Traceability:** The statistically significant reduction in Precision@3 under the stricter 2-1-0 relevance rule ($p = 0.0002$) shows that many highly ranked candidates that are semantically relevant are not necessarily judged by the SME to represent full realization of the source element. In the evaluated architecture, this distinction was particularly relevant because architectural elements may have partial or overlapping relationships. The finding supports positioning the proposed workflow as a decision-support mechanism that prioritizes candidate relationships for engineering review rather than as an autonomous traceability classifier. SME assessment remains necessary in the evaluated workflow to distinguish accepted relationships from partial or rejected mappings.

V. Conclusion

This study investigated an LLM-assisted semantic traceability workflow for establishing mappings across DoDAF architectural viewpoints. The proposed approach combines dense semantic embedding retrieval with cross-encoder re-ranking and evaluates the resulting candidate rankings against SME judgments using rank-aware Information Retrieval metrics. The workflow was evaluated using a real-world PNT Enterprise Architecture containing Capability, Operational, and Service viewpoints, providing an empirical setting for examining automated cross-viewpoint mapping in an MBSE context.

Across the three mapping tasks evaluated—Capability-to-Operational Activity, Operational Activity-to-Service, and Capability-to-Service—the workflow consistently identified relevant candidates within a top-3 review budget. Success@3 was 1.000 for the Capability-to-Operational Activity and Capability-to-Service mappings and 0.950 for Operational Activity-to-Service. These results indicate that semantic retrieval can provide high-yield candidate sets for SME review across the evaluated viewpoints. At the same time, the reduction in Precision@3 under the stricter 2-1-0 relevance rule demonstrates that semantic similarity does not, by itself, reliably distinguish complete realization from partial or overlapping functional relationships. The SME therefore remains an important component of the traceability process, particularly where architectural relationships involve multiple complementary elements.

The comparison between cosine-based retrieval and cross-encoder re-ranking also provides a more nuanced result than a simple preference for the more complex retrieval strategy. Although re-ranking improved Top-1 Acceptance Rate and MRR@3 for Operational Activity-to-Service mapping, the observed differences were not statistically significant. For the two capability-oriented mappings, cosine retrieval produced slightly higher Top-1 and MRR@3 values, also without statistically significant differences. Because re-ranking was applied only to candidates already retrieved in the initial top-3 set, the two methods produced identical Success@3 and Precision@3 values. Within the scope of this case study, these results therefore do not establish a general performance advantage for cross-encoder re-ranking over single-stage semantic retrieval.

The primary implication for systems engineering practice is that LLM-based semantic retrieval can serve as a practical decision-support mechanism for cross-viewpoint traceability, while responsibility for accepting and interpreting architectural relationships remains with systems engineering experts. The results further suggest that the value of such workflows should be assessed not only by whether they retrieve semantically related elements, but also by how well they support engineers in distinguishing fully realized relationships from partial or supporting relationships.

Building upon the findings of this study several key avenues for future investigation emerge:

1) **Expanded DoDAF Viewpoint and Metamodel Coverage:** Future work will extend the semantic workflow to additional DoDAF or UAF viewpoints and incorporate structural metamodel constraints directly into the candidate retrieval process to prevent semantically plausible but structurally invalid mappings.
2) **Expanded Coverage Across Enterprise Architectures:** While this study focuses on a single PNT EA, future work should extend the workflow to support cross-enterprise and cross-organizational mapping, facilitating model reuse, cross-domain interoperability, and broader architectural alignment.
3) **Direct Cameo Toolchain Integration:** To further streamline systems engineering workflows, future efforts will focus on embedding the two-stage AI retrieval pipeline into open-source Cameo Systems Modeler plugins, enabling automated, real-time traceability checks directly inside the modeling environment.
4) **Fine-Tuning on Domain-Specific Systems Engineering Corpora:** Investigating parameter-efficient fine-tuning (PEFT) on specialized defense, aerospace, and MBSE datasets may further refine cross-encoder re-ranking accuracy across domain-specific jargon and technical abbreviations.

Finally, it is important to point out that this study is limited to a single PNT Enterprise Architecture, three DoDAF

viewpoints, the selected embedding and re-ranking models available on the secure Artificial Intelligence Platform, and validation based on SME judgments. Accordingly, the findings should be interpreted as evidence from an industrial case study rather than as a general demonstration of LLM-based traceability across all enterprise architectures or MBSE frameworks.

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